

Security Report Task 1

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Domain: Cyber Security

Task Name: Web Application Security Testing

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TOOLS USED

- Kali Linux
- OWASP ZAP
- DVWA
- Burp Suite

OBJECTIVE

To perform web application security testing and identify common vulnerabilities like SQL injection, XSS, and Brute Force.

RESULTS

1) SQL Injection:

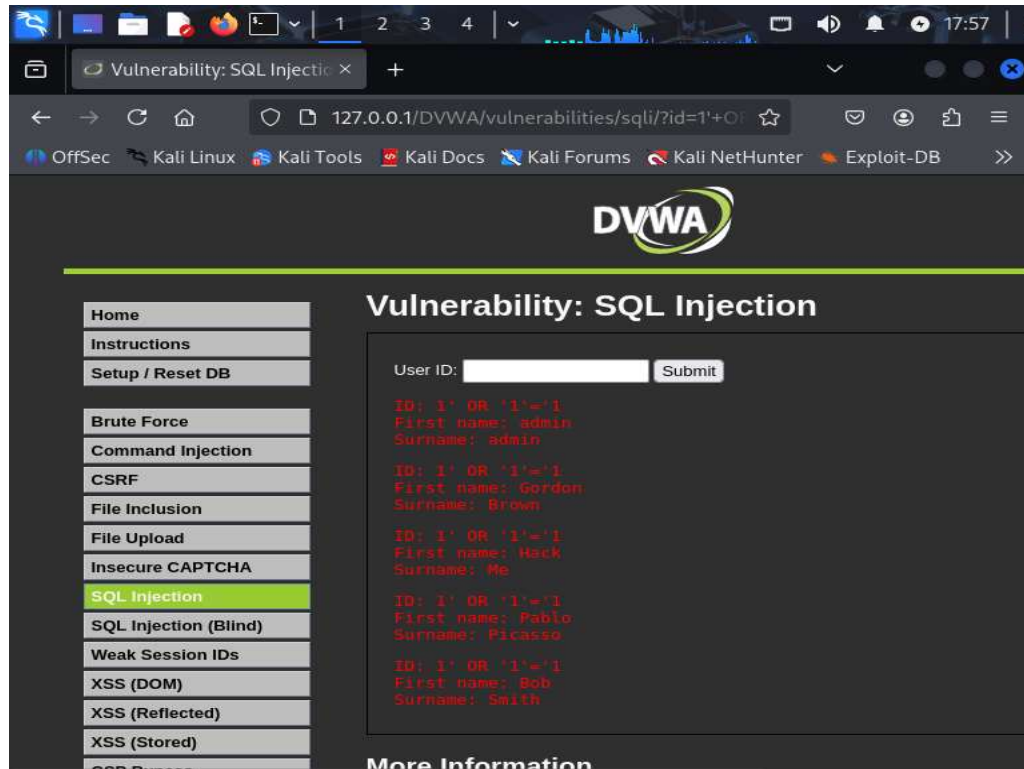
Input: 1' or '1'='1

Outcome: Logged in successfully without valid credentials.

Suggestions:

- Escape user inputs properly before using them in queries.
- Use prepared statements with parameterized queries.

Screenshot:



2) Cross-Site Scripting (XSS):

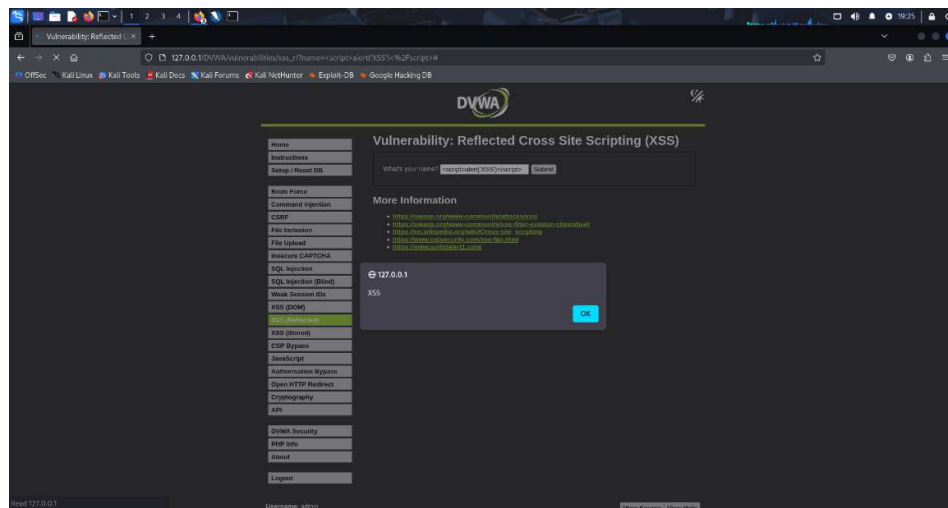
Input: `<script>alert('XSS')</script>`

Outcome: Input was reflected without sanitization confirming Reflected XSS vulnerability.

Suggestions:

- Validate and sanitize all user inputs.
- Encode output before rendering in HTML/JS contexts.
- Use modern frameworks with built-in XSS protection.
- Implement a strong **Content Security Policy (CSP)**.

Screenshot:



3) Brute Force:

Input: Username = 'admin'

Password = 'password'

Outcome: Successfully logged in by brute forcing the username and password.

Suggestions:

- Enforce strong password policies.
- Implement account lockout or rate limiting after failed attempts.
- Use CAPTCHA or multi-factor authentication (MFA).

Screenshot:

